Assessing POD Controller Robustness via Eigenvalue Sensitivities

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***Abstract*—Guaranteeing sufficient damping of inter-area oscillations is a critical requirement for Transmission System Operators (TSOs). The increasing penetration of power electronics–interfaced devices (PEIDs) introduces significant variability in operating conditions, complicating the design of robust Power Oscillation Damping (POD) controllers.**

This paper presents a novel framework for assessing POD robustness using symbolic computation of eigenvalue sensitivities (ES). The method provides analytical expressions of ES as explicit functions of system variables for different POD inputs and outputs, enabling systematic evaluation of signal selection across a wide range of operating points. The approach is developed on a Two-Machine System, and the results are applied to the Two-Area Four-Generator System.

***Index Terms*—Inter-area Oscillations, POD design, Robustness, Symbolic Analysis.**

I. Introduction

The rapid evolution of power systems, driven by the energy transition and the integration of energy markets, has led to increasingly variable operating conditions. These changes have a pronounced impact on inter-area oscillations, small-signal stability phenomena in which groups of synchronous generators in different areas swing against each other. Such oscillations can restrict power transfer among regions and increase operational costs. Power electronics–interfaced devices (PEIDs), which lack rotating masses or decouple them from the grid, do not inherently participate in these oscillations when operating under grid-following mode. However, when equipped with power oscillation damping (POD) controllers, they can actively contribute to damping. The POD must perform adequately under different operating conditions, a requirement referred to as POD robustness.

Robustness can be achieved either by finding robust but fixed POD parameters or by adapting the parameters to the system in real time. Adaptive control approaches have been explored, for instance in [1] a phase-compensation adaptive tuning for PV is proposed. Nonetheless, fixed-parameter approaches remain the preferred choice for transmission system operators (TSOs), as they are easier to apprehend, implement in simulations, and incorporate during the conception and design stages. This preference justifies the focus of this paper on fixed-parameter POD designs.

The robustness of fixed-parameter PODs has been addressed in several studies. Several works consider variations in generation dispatch; for instance [2] addresses the tuning of a lead–lag POD for wind power plants (WPPs), whereas [3] focuses on PODs for battery energy storage systems (BESSs) and photovoltaic (PV) plants. The robustness of a POD is strongly influenced by the selection of its input–output (I/O) signals. Previous studies have explored this aspect; for instance, [4] discusses I/O selection while tuning a lead–lag POD for WPPs on a benchmark system.

Numerous techniques have been proposed in the literature to tune PODs, but two are mainly used in practice: frequency-domain and eigenvalue analysis, and eigenvalue sensitivity (ES) methods. The frequency domain and eigenvalue analysis method is well summarized in [5]. ES have also been successfully used for POD tuning, including FACTS devices [6] and HVDC (High-Voltage Direct Current) links [7].

This paper proposes a novel approach to enhancing POD robustness through systematic analysis of the input-output (I/O) signals of the POD. The proposed method aims to identify I/O pairs that ensure robustness across a wide range of operating points, defined to include variations in often-overlooked variables such as inertia imbalance, power flow and impedance between oscillating regions, and the voltage at the point of connection of the POD. The approach builds on the symbolic computation of ES, which are well established as an effective tool for both small- and large-scale systems due to their straightforward computation. However, ES have not yet been exploited to address robustness. By leveraging symbolic computation [8], analytical expressions of the ES are obtained as explicit functions of system variables. This provides deeper insight into system behavior and facilitates the identification of robust I/O choices, in contrast to numerical methods, where

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gaining a comprehensive view of system dynamics over a wide and continuous range of operating points is more difficult. The effectiveness of the proposed framework is demonstrated on the Two-area Four-Generator System.

The paper is organized as follows. Section II introduces the concept of ES and justifies their use for assessing POD robustness. Section III describes the symbolic procedure used to compute ES and its applicability for designing control signals. Section IV presents the results obtained from the symbolic analysis of ES. Section V applies these results to the Two-Area Four-Generator System, demonstrating their relevance. Finally, Section VI concludes the paper.

II. Eigenvalue Sensitivities for Robust POD I/O Selection and Tuning

Designing a POD requires careful consideration of both the I/O selection and the POD tuning. ES can be employed to this end, as they provide direct information on how a given eigenvalue will move when a POD, tuned for a specific I/O pair, is applied to the system. While the magnitude of ES indicates the capacity of the POD to shift the eigenvalue, its phase indicates the direction of the shift. Eigenvalue trajectories are generally nonlinear [9]; therefore, ES provide only a first-order approximation that is valid in the vicinity of the eigenvalue, as illustrated in Figure 1.

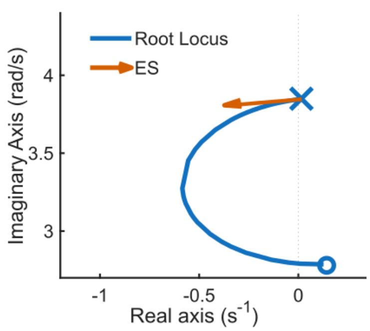

Fig. 1. Eigenvalue trajectory (root locus) and ES.

Formally, the ES S_i^k of the i-th eigenvalue λ_i with respect to the controller gain k is defined as:

$$S_i^k = \frac{\partial \lambda_i}{\partial k} \tag{1}$$

A. ES for Robust POD I/O Selection

In this paper, the scope of I/O selection is extended by introducing an additional requirement: across multiple operating points, the ES magnitude must not only be high to ensure POD effectiveness, but the ES must also maintain a consistent orientation. This guarantees that the POD will not adversely affect system stability under varying operating conditions when a common compensation network is applied. Ideally, ES phases across different operating points should generally stay within about $90°$ of each other, which translates to a maximum deviation of roughly $45°$ around the horizontal axis in the compensated response.

In the typical case where the POD employs a lead–lag structure, its primary effect is to rotate the ES vector by a constant phase shift, aligning it towards the left half of the complex plane. When ES maintain a consistent orientation across different operating points, the common phase shift introduced by the controller remains effective in all cases and does not compromise system stability. This geometric interpretation is illustrated in Figure 2.

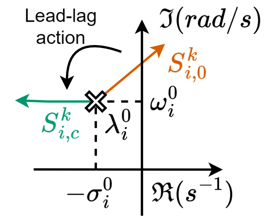

Fig. 2. Geometric interpretation of the eigenvalue sensitivity, where $S_{i,0}^k$ is the initial ES and $S_{i,c}^k$ is the compensated ES.

B. ES for Robust POD Tuning

In this work, the robust tuning method proposed in [10] is employed to design a lead–lag POD capable of damping inter-area oscillations within a specified frequency range across multiple operating points. As will be demonstrated in the example presented later, the effectiveness of this method depends critically on the appropriate selection of I/O signals.

The structure of the POD is illustrated in Figure 3, which can include one to three lead-lag compensation networks.

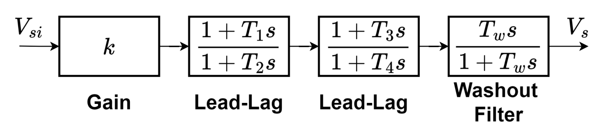

Fig. 3. Control Scheme of the POD.

This method aims to achieve the desired damping ratio for the inter-area mode, λ, across a set of N_{op} operating points. The algorithm consists of two main steps: the design of the lead–lag filter and the design of the stabilizer gain.

1) Design of the Lead-lag Filter: The parameters of the lead–lag filter, T_1 and T_2, are obtained by solving a nonlinear optimization problem to align the phase of the ES of the inter-area mode as close as possible to $180°$ at all N_{op} operating points. The problem is formulated as:

$$\begin{array}{ll} \text{Find:} & T_1 \\ \text{and solve:} & \min \sum_{j=1}^{N_{\text{op}}} \gamma_j \cos\left(\arg(S_j(T_1))\right), \\ \text{where:} & \gamma_j \text{ is a weighting factor} \propto |S_j(T_1 = 0)|, \\ & S_j \text{ denotes the ES at the } j\text{-th OP.} \\ \text{Set:} & T_2 = \dfrac{T_1}{\alpha}, \quad \alpha \text{ computed from } S_j(T_1 = 0). \end{array} \tag{2}$$

2) Design of the Stabilizer Gain: The POD gain k is determined by solving a linear optimization problem to ensure that the estimated damping ratio of the inter-area mode meets or exceeds the desired value ζ_d at all N_{op} operating points considered. The problem is formulated as follows:

$$\begin{aligned}
&\text{Find:} && k \in (0, k^{\max}) \\
&\text{such that:} && k\,\Re\!\left(\frac{\partial \lambda_j}{\partial k}\right) \le \Re\big(\lambda_j^d - \lambda_j^0\big), \quad j = 1, \ldots, N_{\text{op}}, \\
&\text{where:} && \lambda_j^d = -\zeta_d\,\Im\big(\lambda_j^0\big) + j\,\Im\big(\lambda_j^0\big), \quad j = 1, \ldots, N_{\text{op}}, \\
&\text{and solve:} && \min k.
\end{aligned} \tag{3}$$

Here, λ_j^0 denotes the eigenvalue of the inter-area mode at operating point j, and λ_j^d is the corresponding eigenvalue associated with the desired damping ratio ζ_d. The optimization yields the minimum POD gain k that guarantees the required damping across all considered operating points.

III. Symbolic Robustness Analysis

A. Symbolic Computation of Eigenvalue Sensitivities for a Two-Machine System

The symbolic computation of the ES S of the i-th eigenvalue, for a given I/O selection i/o with respect to the POD gain k, enables a deeper understanding of how these ES depend on the system variable v that defines the operating points j. The ES is expressed as a function of these parameters:

$$S^k_{i,i/o} = f(k, v), \quad v \in \mathcal{V} \tag{4}$$

$\mathcal{V}$ denotes the set of system variables considered in the symbolic analysis, as introduced in Section III-B. The parameters k and v are defined symbolically throughout the study.

Symbolic computation provides exact, analytical representations of system equations, enabling closed-form expressions for ES as explicit functions of v and k. It allows systematic exploration of variable ranges and offers deeper insight for control design. However, symbolic methods are limited by system size, equation complexity, and computational resources. Consequently, they are most useful as a diagnostic and design tool, with results validated numerically on larger systems.

In this study, symbolic computation is applied to a simple Two-Machine System, which is a reduced version of the Two-Area Four-Generator System presented in [5]. The synchronous machines are modeled using the classical two-axis model, which captures the essential electromechanical dynamics relevant for small-signal stability analysis. A shunt device is intended to be connected at bus 3 representing a generic PEID. It is located between the two synchronous machines, at a point twice as far from generator 2 as from generator 1. The objective is to equip this device with a POD that modulates its power injection to mitigate the system's inter-area oscillation.

The static parameters of the two-machine benchmark system are summarized in Table I. The dynamic parameters are the same as in [5], except that the inertia constant of both

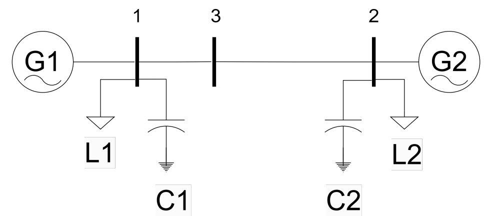

Fig. 4. Two-machine System.

machines is set to $H = 6.5\ MW \cdot s/MVA$. The system configuration is shown in Figure 4. Modal analysis identifies a dominant electromechanical inter-area mode at 0.52 Hz.

TABLE I
STATIC PARAMETERS OF THE TWO-MACHINE SYSTEM.

Bus	P_L [MW]	Q_C [Mvar]	P_G [MW]	Q_G [Mvar]	V_m [p.u.]
1	1767	100	1388	404	1.03
2	967	100	1388	404	1.03
3	0	0	0	0	1.00

Branch	From–To	r [p.u.]	x [p.u.]	b [p.u.]
1	1–3	0.007	0.070	0.1925
2	3–2	0.015	0.150	0.1925

The steps for computing the symbolic ES are summarized as follows:

1) Perform a power flow calculation to obtain the complete set of operating point variables for system linearization.
2) Obtain the open-loop multimachine state-space representation following the procedure described in [5], obtaining the state matrix $A(v)$.
3) Obtain the controllability vector $b(v)$ and observability vector $c(v)$.
4) Form the closed-loop system with input u and output y using a proportional feedback gain k, defined by $u = -ky$ and $A_{\text{cl}}(v,k) = A(v) - b(v)kc(v)$.
5) Derive the characteristic equation of the closed-loop system as $\det(A_{\text{cl}}(v,k) - \lambda I) = 0$.
6) Compute the symbolic eigenvalues $\lambda_i(v,k)$ analytically by solving the characteristic equation. The equation is analytically solvable because its degree is at most 4, as the classical synchronous machine model is used. According to the Abel–Ruffini theorem, only polynomial equations of degree up to four can be solved analytically in terms of radicals.
7) Differentiate $\lambda_i(v,k)$ with respect to k to obtain the symbolic expression of ES $S_i(v,k)$. This expression is then evaluated at $k = 0$, as the open-loop ES is used for POD design.
8) Once the sensitivity is obtained as an expression of the parameter of interest, $S_i(v)$ its robustness can be analyzed under different operating conditions, defined by variable v.

B. Variables Considered in Symbolic Analysis

The set of variables $\mathcal{V}$ considered in the symbolic analysis includes the inertia ratio between rotating masses, network impedance, voltage magnitude at the point of connection of the POD, and the system power flow pattern. The objective is to assess a range of operating conditions resulting from factors such as increasing renewable energy integration, variations in generation dispatch, transmission line maintenance, and seasonal effects on voltage levels. The studied ranges for each variable are summarized in Table II.

The inertia ratio between rotating masses is obtained by fixing the inertia of one machine and varying that of the second as a multiple of the first. The network impedance is determined by adjusting the impedance of the line connecting buses 2 and 3. The power flow pattern is quantified through the voltage angle, as expressed by the active power transfer equation:

$$P_{ij} = \frac{V_i V_j}{x_{ij}} \sin(\delta_i - \delta_j) \tag{5}$$

where V_i and V_j are the voltage magnitudes at buses i and j, respectively; δ_i and δ_j are their corresponding phase angles; and x_{ij} is the series reactance between buses i and j.

TABLE II
RANGES OF VARIABLES CONSIDERED IN SYMBOLIC ANALYSIS

Variable	Range	Units
Inertia ratio	[0.1, 10]	–
Line Impedance	[0.015, 1.5]	p.u.
Power flow Pattern (voltage angle)	[0, 2π]	rad
Voltage magnitude	[0.5, 1.5]	p.u.

C. I/O Considered in Symbolic Analysis

The symbolic calculation of ES is performed for various I/O combinations, with the analysis restricted to single-input single-output (SISO) systems and a PEID type shunt. The I/O considered are listed in Table III. All inputs are based on local measurements.

TABLE III
INPUTS AND OUTPUTS FOR A POD OF A SHUNT PEID

Inputs	Bus Frequency Bus Voltage Line Current Line Active Power Line Reactive Power Line Apparent Power
Outputs	Active Power Injection Reactive Power Injection

D. Input Design Based on Symbolic ES

In a linear system, an output defined as a linear combination of the entries in Table III yields an ES that is the same linear combination of the ES corresponding to the individual signals. This concept is illustrated in Figure 5.

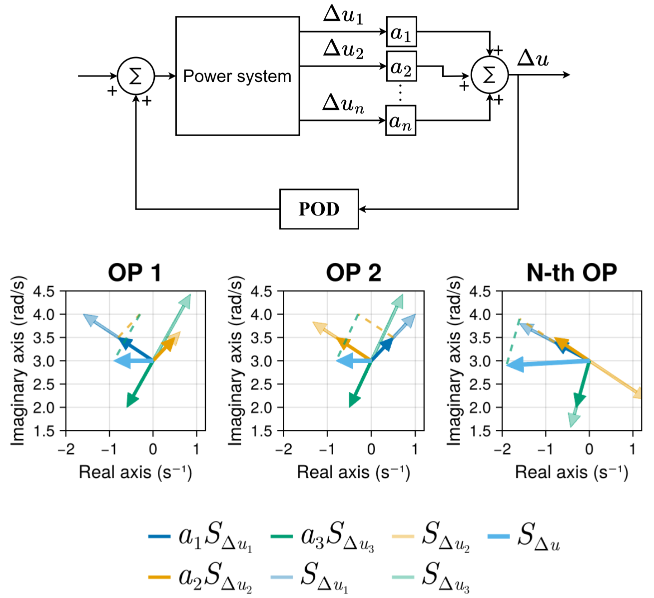

Fig. 5. Illustration of the signal construction process, aiming to determine a common coefficient vector $\boldsymbol{a}$ that ensures alignment of the resulting ES $S_{\Delta u}$ across operating points (OP), calculated as a linear combination of the original ES ($S_{\Delta u_n}$)

To identify a combination of POD inputs whose ES preserves a consistent direction across all operating points, the following linear program is formulated:

$$\begin{aligned}
&\text{Given:} && \mathbf{g}(v) = \left[S(v,\mathbf{u}_1) \quad S(v,\mathbf{u}_2) \quad \cdots \quad S(v,\mathbf{u}_m)\right]^\top, \\
&\text{Find:} && \boldsymbol{a} = [a_1, a_2, \dots, a_m]^\top \text{ and } t \geq 0 \\
&\text{such that:} && \mathbf{g}(v)^\top \boldsymbol{a} \geq t, \quad \forall v \\
& && -1 \leq a_j \leq 1, \quad j = 1, \dots, m, \\
&\text{and solve:} && \max_{\boldsymbol{a},\, t} \ t.
\end{aligned} \tag{6}$$

Here, $\mathbf{g}$ collects the ES values for all operating points and candidate input signals. Maximizing the minimum value t yields coefficients $\boldsymbol{a}$ that ensure the combined ES remains positive. Since the analysis is based on the classical model, the ESs are purely imaginary eigenvalues, which provide an approximation of the oscillatory behavior of a more complex system.

The objective of this procedure is to derive robust POD input signals that can effectively damp inter-area oscillations in a generalized manner. The underlying oscillation mechanism—active power exchange between distant synchronous generators—remains fundamentally the same regardless of system size. The results presented here are obtained using a reduced Two-Machine System, which captures the essential structural behavior of inter-area oscillations. It is important to note that the optimization of inputs can be performed even with only two operating points; however, this approach risks missing important information about other potential operating conditions. Symbolic analysis allows exploration across multiple operating points, providing a more comprehensive assessment of POD robustness.

IV. Results of the Symbolic Analysis

At the linearization point, all eigenvalues are purely imaginary due to the absence of damping, and the corresponding ES are also purely imaginary. The only exception is frequency, whose ES are purely real. This occurs because frequency is computed as a time derivative, introducing a 90° phase shift in the response. Consequently, to ensure robustness, the ES must remain consistently positive or negative for a given I/O pair.

The resulting eigenvalue frequencies and their corresponding ES are presented in Figures 6 and 7, respectively. When frequency is used as the POD input, the ES exhibits the same qualitative behavior, and thus the conclusions remain unchanged. A sample of these results is shown separately in Figure 8 due to their smaller order of magnitude. The equations obtained from the symbolic analysis that are used to generate the figures are available in Appendix A. When computing the system model, the inertia ratio and line impedance primarily affect the system matrix $A(v)$, whereas voltage magnitude and power flow have a stronger impact on observability and controllability factors.

Figure 6 shows the variation of the inter-area mode frequency. The variable ranges have been normalized for direct comparison within the same figure. The inter-area mode frequency remains unchanged regardless of the I/O pair, as it is an inherent property of the system's stability. The **line reactance** has the strongest influence on the eigenvalue frequency, reducing the mode frequency as the electrical distance between generators increases, as expected. In contrast, **power flow pattern** and **voltage magnitude** have negligible effect. Finally, a slight variation is observed with the **inertia ratio**, although this appears to depend more on the system's total inertia than on the ratio itself.

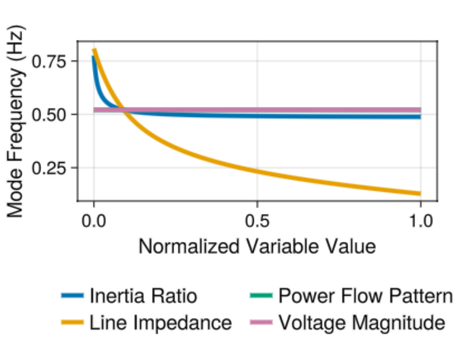

Fig. 6. Inter-area mode Frequency as a Function of System Variables

ES analysis highlights the strong influence of the operating point on the effectiveness of POD. Variations in the **inertia ratio** have a pronounced and generally monotonic effect: for both active and reactive power injection, the ES changes sign beyond a certain threshold. This monotonic behavior allows for the definition of clear validity limits for the POD design. The influence of **line impedance** is less predictable. Although sign preservation cannot be ensured over the entire impedance range, the ES evolves smoothly, indicating that robustness is likely maintained for most practical operating conditions. The greatest challenge arises from variations in **power flow patterns**. As shown analytically in Appendix A, the ES expression consists of multiple sinusoidal components, making every I/O combinations exhibits oscillatory sign changes. Power flow is the most challenging variable and can easily change the sign of ES depending on the operating point. In contrast, **voltage magnitude** primarily affects the magnitude of ES rather than its sign, except when reactive power is used as the input signal. Consequently, robustness issues related to extreme voltage levels can be largely ruled out.

Overall, robustness cannot be guaranteed for any I/O pair across all operating conditions, mainly due to the dependence on power flow. Several ES trajectories change sign as system variables vary, implying that the POD contribution may alternate between stabilizing and destabilizing effects.

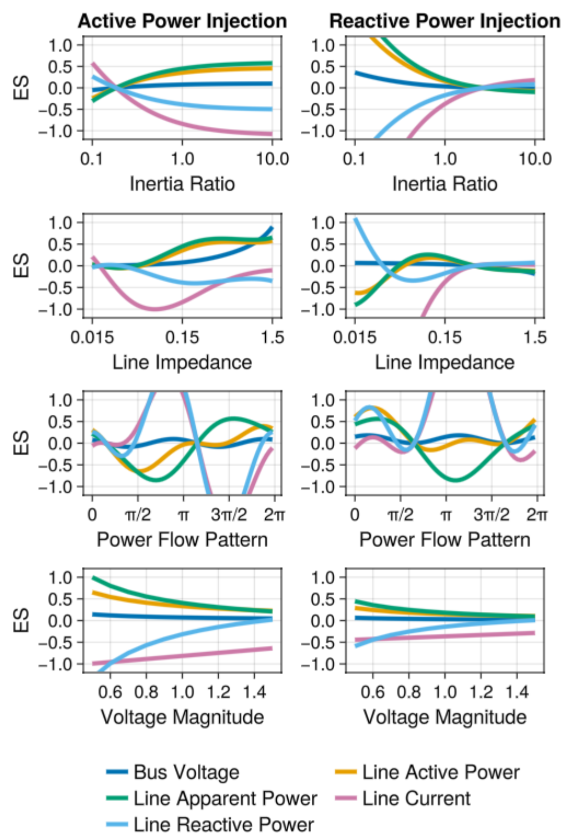

Fig. 7. Eigenvalue Sensitivities (ES) as a Function of System Parameters

A. Example of Input Design Based on Symbolic ES

Based on the results obtained in the previous section, the solution proposed in this paper for designing a fixed-parameter POD that ensures stability across a wide range of operating

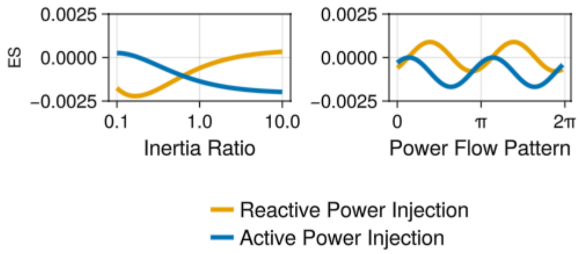

Fig. 8. Eigenvalues Sensitivities (ES) when Using Frequency as Input

conditions consists of defining the POD input as a linear combination of other input signals. In this paper, a specific application is considered in which the POD output corresponds to either active or reactive power injection (POD-P or POD-Q, respectively), while the input is defined as a linear combination of the bus voltage and the line active and reactive power flows. It should be noted that other signal combinations could also be used to achieve similar objectives. The resulting input coefficients, obtained by solving a linear programming (LP) problem, are presented in Table IV. The resulting coefficients are applied to the Two-Area Four-Generator system to verify the scalability of the proposed control scheme and its applicability to larger systems.

TABLE IV
COEFFICIENTS **A** OBTAINED FROM THE LP PROBLEM

Input Signal	POD-P	POD-Q
Bus Voltage	-1	-1
Line Active Power	0.1659	0.0726
Line Reactive Power	0.0411	0.0180

V. Application to the Two-Area Four-Generator System

The Two-Area Four-Generator System is used to illustrate the ideas discussed in the preceding sections. The parameters of the system can be found in [5]. Four operating points are examined, as summarized in Table V. Three electromechanical oscillation modes can be identified in the system: two local modes and one inter-area mode. The inter-area mode has negative damping indicating an unstable system. A POD of a BESS will be tuned to improve the damping of the inter-area oscillation. The BESS is modeled by a simplified first-order transfer function, representing common outer active and reactive power controls. Inner control loops are not represented for having a minor impact on the electromechanical oscillations. Three potential locations are considered: bus 7, bus 8 and bus 9. Although PODs connected closer to generators are generally more effective, the goal is to place the BESS and its POD at non-generator buses within the transmission system to explore new device placements. In this study, we assume the BESS neither produces nor absorbs energy in the steady-state operation, and it only injects or absorbs energy when required by the POD.

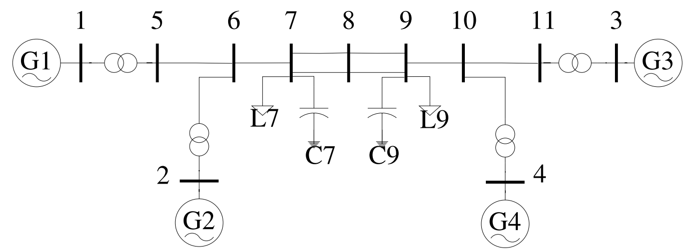

Fig. 9. Two-area Four-Generator system.

TABLE V
OPERATING POINTS (OP) OF THE TWO-AREA FOUR-GENERATOR SYSTEM
OOS STANDS FOR "OUT OF SERVICE"
DAMPING AND FREQUENCY REFER TO THE INTER-AREA MODE.

OP	*L7*	*L9*	Line OOS	Damping	Frequency
1	967 MW	1767 MW	None	-0.16%	0.61 Hz
2	1767 MW	967 MW	None	-0.93%	0.60 Hz
3	967 MW	1767 MW	Bus 8 - 9 (1)	-0.73%	0.46 Hz
4	1767 MW	967 MW	Bus 7 - 8 (1)	-2.21%	0.41 Hz

In this example, we first attempt to tune a POD, either POD-P or POD-Q, using voltage as the input. The ES are calculated, showing that, for both active and reactive power injections, they are misaligned across the four OPs. This indicates that no robust POD can be obtained using voltage as the input. As shown by previous results, this behavior is likely to occur for any other input proposed in Table III.

Consequently, we apply the LP-optimized inputs obtained in Section IV-A. Figure 10 compares the ES for voltage and for the LP-optimized inputs. The ES represented with lighter (semi-transparent) colors systematically point in different directions, particularly at buses 7 and 9, where their magnitude is higher and the POD can have a larger impact. In contrast, when the LP-optimized input is used, the ES are consistently aligned; that is, their phase angles remain within a 90° interval across all operating points, thereby enabling the design of a POD that is effective for all OPs.

Based on these results, both POD-P and POD-Q are tuned for bus 7. For buses 8 and 9, the sensitivities are similarly aligned; however, their magnitudes are small, indicating limited effectiveness of the POD. While a low ES magnitude does not compromise system stability, it may reduce the POD's impact. Therefore, bus 7 is selected as the most suitable installation point.

The results are shown in Figure 11, where the original eigenvalues are compared with their positions after applying the PODs. As expected, POD-P outperforms POD-Q.

VI. Conclusions

This paper proposes a novel approach based on symbolic ES, enabling a detailed assessment of the robustness of PODs for damping inter-area oscillations. The analysis highlights that the most challenging factor for ensuring adequate damping is the change in power flow patterns. It also raises awareness of other power system variables that can affect POD effectiveness on some occasions, such as variations in inertia or in the electrical distance between rotating masses.

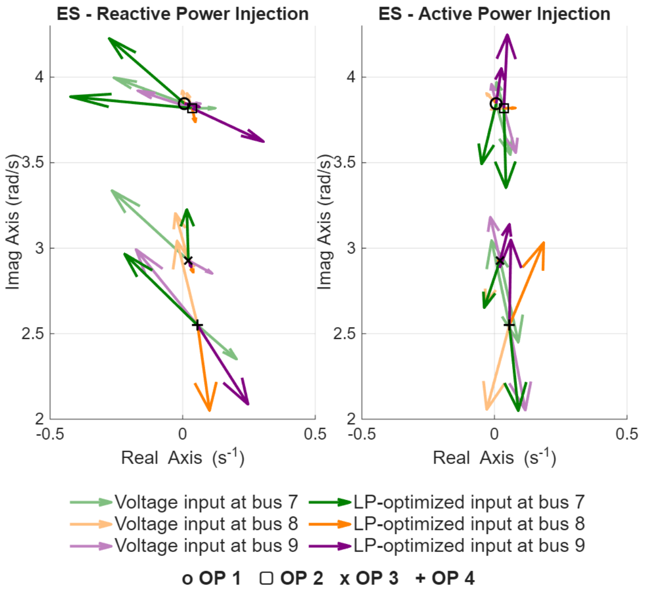

Fig. 10. Comparison of the inter-area mode ES for reactive (POD-Q) and active (POD-P) power injection.

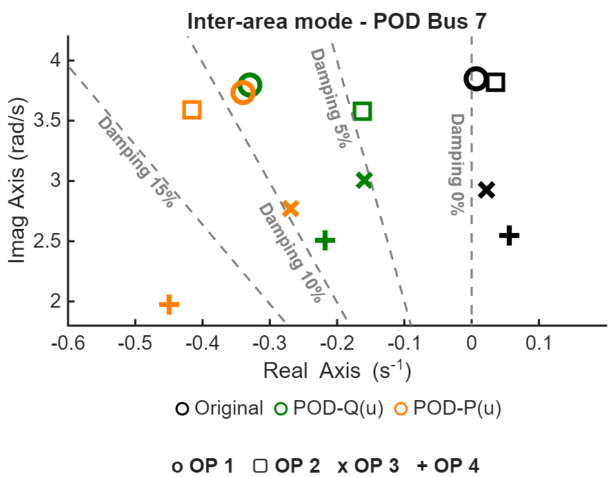

Fig. 11. Eigenvalue Results with POD-Q and POD-P for LP-optimized input u.

This study was not able to identify an I/O combination that is robust across all operating points studied. Therefore, a method to obtain robust POD inputs is proposed. The procedure relies on computing these inputs by solving a LP problem on a reduced system. The LP-optimized inputs are applied to the Two-Area Four-Generator System, showing promising results, however further research is required to confirm its effectiveness and applicability to larger-scale systems.

Appendix

The following expressions are given as functions of the coefficients α.

$$\alpha_k \in \begin{cases} \mathbb{R}, & \text{for frequency input,} \\ i\mathbb{R}, & \text{for other inputs.} \end{cases} \tag{7}$$

where the coefficients α_k take different values and assign different weights to the terms, depending on the chosen I/O.

- Power Flow Pattern ($\mathtt{v}_1$) :

$$\begin{aligned} S_{v_1} = (&\alpha_1 \sin(2\mathtt{v}_1) + \alpha_2 \sin(\mathtt{v}_1) + \alpha_3 \cos(\mathtt{v}_1) \\ &+ \alpha_4 \sin^2(\mathtt{v}_1) + \alpha_5 \cos^2(\mathtt{v}_1)) \end{aligned} \tag{8}$$

- Voltage magnitude ($\mathtt{v}_2$):

$$S_{v_2} = \frac{\alpha_1 + \alpha_2 v_2}{v_2^{\beta}}, \quad \beta = \begin{cases} 2 & \text{for frequency input,} \\ 1 & \text{for other inputs.} \end{cases} \tag{9}$$

- The analytical expressions for line impedance and inertia ratio are not provided due to their high complexity.

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